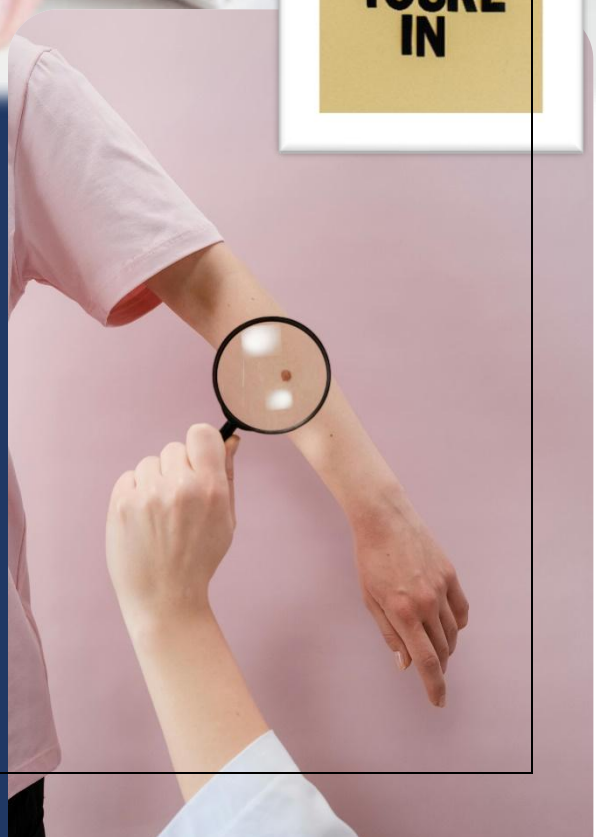
Advanced Analysis and Predictive Modelling for Identifying *Erythemato-Squamous Disease Types*

Data Analysis Project II – Group 4
ST 4052 Statistical Learning II



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Abstract

The goal of this study is to categorize erythemato-squamous diseases (ESDs) using machine learning techniques in the interest of improved diagnosis and clinical decision. The information consists of clinical and histopathological features of six disease classes. A detailed descriptive analysis and cluster analysis were performed to determine patterns and similarities among the classes. FAMD was used for reducing dimensionality, and SMOTE was used for balancing classes to enhance model performance.

Several machine learning algorithms were trained and evaluated, including Naive Bayes, Random Forest, Support Vector Machine (SVM), Ridge, and Lasso classifiers. After comparing the performance metrics, the Decision Tree model performed best with 99% accuracy on the test set. Due to its accuracy rate and interpretability, the Decision Tree model is a suitable classification tool for ESD types. The results demonstrate the ability of machine learning to aid dermatological diagnosis and provide insight into disease patterns.

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Introduction

Differential diagnosis of Erythmato-Squamous Diseases (ESD) is a major challenge in the field of dermatology due to the overlapping symptoms and similar visual manifestations among variance conditions. The ESD diseases are placed into six different classes that are categorized by redness(erythema) and scaling, and they require accurate diagnostic to ensure appropriate treatment.

The clinical importance of distinguishing between these diseases cannot be overstated. Misdiagnosis may lead to inappropriate treatment plans, prolonged patient discomfort, and potentially severe long-term complications. As these conditions can vary in their underlying causes, prognosis, and required therapeutic interventions, a precise understanding of their distinguishing features is essential for effective patient management.

Description of the Question

Building on the results of the descriptive analysis, the advanced analysis phase is focused on applying machine learning techniques to enhance our understanding and predictive information concerning erythemato-squamous diseases (ESDs). The overall goal is to develop models which will accurately classify disease types and uncover more subtle, data-based correlations between symptoms and diagnosis.

The question we are going to answer is:

To what extent do machine learning models improve ESD type classification, and which predictors are most crucial to achieving correct diagnosis?

This central question is further explored through the following sub-questions:

- Which machine learning models provide the greatest predictive accuracy for separating the six ESD classes?
- Does model performance or interpretability gain from applying clustering?
- In what way does model performance vary across clusters or data subsets, and can cluster-wise modeling be employed to improve accuracy?
- Can techniques such as SMOTE prove useful for minimizing class imbalance and improving generalizability of classification models?

With this upgraded stage, we not only aim to improve the precision of automated classification of ESDs but also offer clinical informative insights that can assist dermatologists in early diagnosis, reduce misclassification, and identify subtle trends which are imperceptible through visual inspection or plain statistical summaries.

Description of the Data set

This dermatology dataset contains information on 366 patients with various dermatological diseases, represented by a total of 34 attributes (including the class label). The attributes capture both clinical and histopathological observations.

Total Attributes: 34

Explanatory Attributes: 33

Among them, 31 attributes are ordinal and there are one binary attribute and one discrete numerical attribute.

Target Variable (Class): 1 (categorical with 6 diagnostic classes)

Class Label	Class
1	psoriasis
2	seboreic dermatitis
3	lichen planus
4	pityriasis rosea
5	cronic dermatitis
6	pityriasis rubra pilaris

Type of the Attributes	
Ordinal Numeric Attributes 0 : Not present 1 : Mild presence 2 : Moderate presence 3 : Strong presence	Clinical Attributes: 1: erythema 2: scaling 3: definite borders 4: itching 5: koebner phenomenon 6: polygonal papules 7: follicular papules 8: oral mucosal involvement 9: knee and elbow involvement 10: scalp involvement Histopathological Attributes 11: band-like infiltrate 12: melanin incontinence 13: eosinophils in the infiltrate 14: PNL infiltrate 15: fibrosis of the papillary dermis 16: exocytosis 17: acanthosis 18: hyperkeratosis 19: parakeratosis 20: clubbing of the rete ridges 21: elongation of the rete ridges 22: thinning of the suprapapillary epidermis 23: spongiform pustule 24: munro microabcess 25: focal hypergranulosis 26: disappearance of the granular layer 27: vacuolisation and damage of basal layer 28: spongiosis 29: saw-tooth appearance of retes 30: follicular horn plug 31: perifollicular parakeratosis 32: inflammatory monoluclear inflitrate
Binary Numeric Attribute 1 - any of these diseases has been observed in the family 0 - otherwise	Family History
Discrete Numerical Attribute	Age

Table 1: The description of the data set

Main results of the Descriptive Analysis

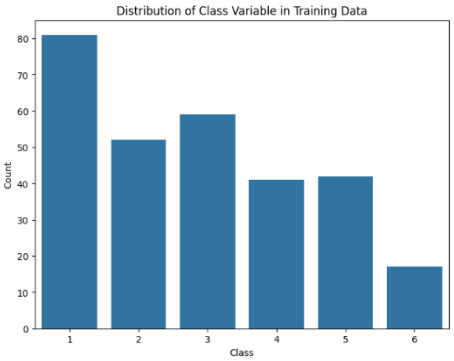


Figure 1: Distribution of the Classes

Class	
1	81
2	59
3	52
4	42
5	41
6	17

Table 2: frequency table of class

Figure 1 shows how the response variable (six diagnostic classes) is distributed in the dataset. Majority of the patients belongs to class 1 which is psoriasis.

By table 2, since class 6 has only 17 observations while class 1 has 81 observations that could lead to class imbalance. This significant difference in the number of samples per class can affect the performance of machine learning models, as they might be biased towards the majority class.

- Kendall's Tau correlation test was chosen as most of the variables of the dataset are ordinal. The heatmap in figure 2, interprets that some of the ordinal explanatory variables are highly correlated indicating potential multicollinearity.

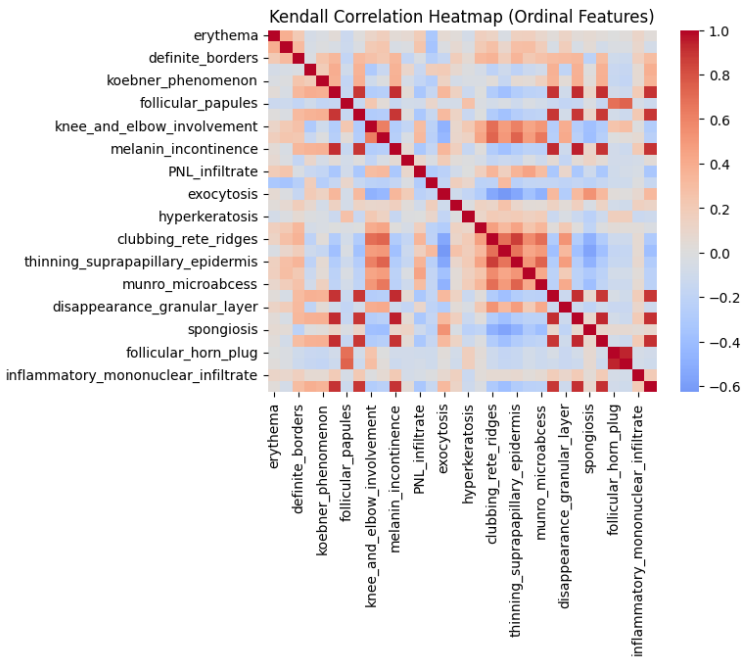


Figure 2: Kendall Correlation Heatmap

Factor Analysis for Mixed Data (FAMD)

The dermatology dataset has 31 ordinal features and one binary response and one discreate numerical variable. We tried to apply FAMD dimension reduction technique to handle mixed data. According to the initial output from the FAMD score plot (figure 5) which indicates clearly 3 clusters of the observations based on the first 2 factor components.

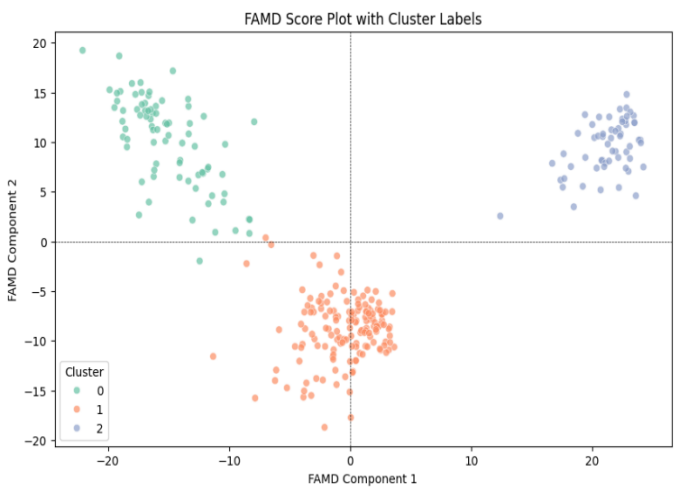


Figure 3: FAMD score plot with cluster labels

Those 3 clusters are further validated by figure 7. The Elbow Method plot shows a clear bend at k = 3, indicating it as an optimal number of clusters where inertia significantly decreases. The Silhouette Score also peaks at k = 3 (>0.75), supporting these three clusters are well-separated.

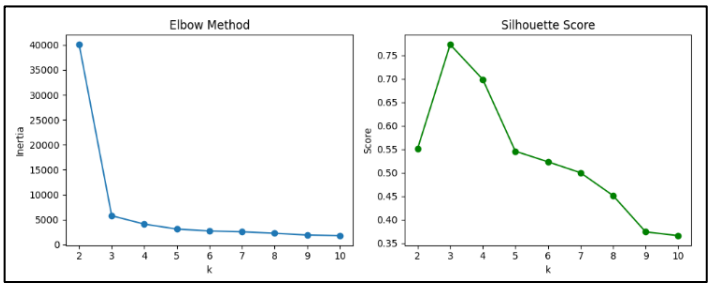


Figure 4: Elbow method plot and Silhouette plot

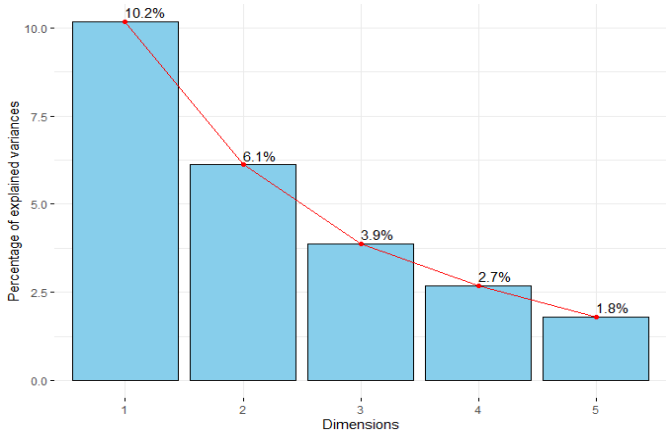


Figure 5: FAMD scree plot

- From the scree plot, it can be observed that the first five dimensions only explain 24.8% percentage of total variation in the dataset. So that important information may be hidden in higher order dimensions.
- By the Loading plot most of the variables contribute to higher dimensions and that also validates that most of features positioned closely together tend to be positively correlated.
- Rather than exploring dimensions, we did cluster analysis with Kmeans clustering technique with FAMD.

Cluster-wise Descriptive Analysis

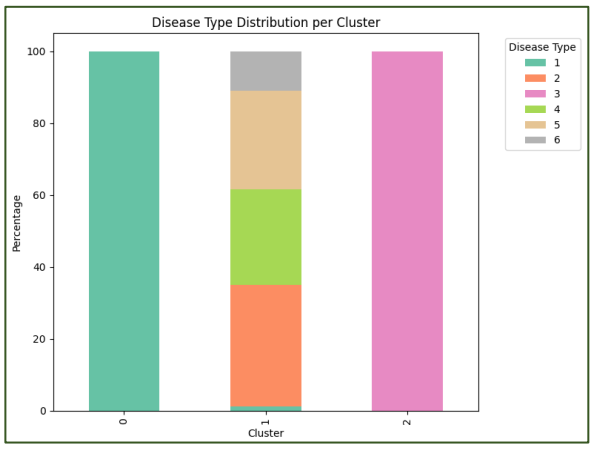


Figure 6: Cluster-wise distribution of diseases

Cluster	No. of observations
0	79
1	154
2	59

Table 3: Frequency Count of clusters

Disease Type	No. of cases
1	2
2	52
4	41
5	42
6	17

Table 4: Disease type distribution in cluster 1

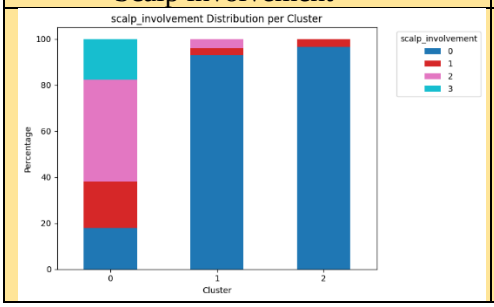
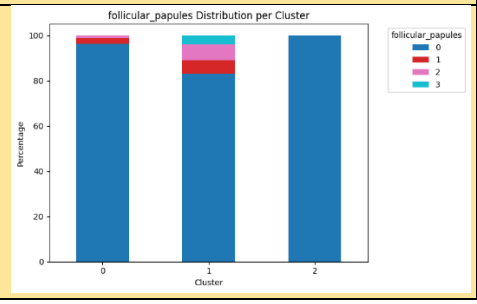
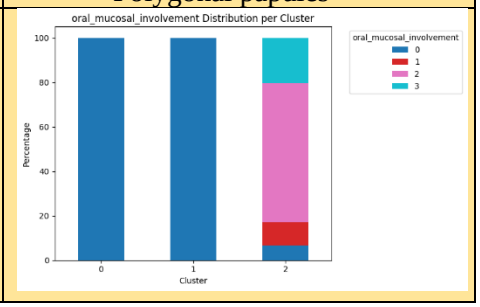
➢ Figure 6 illustrates how disease types are distributed across the clusters. It clearly shows that cluster 0 includes only the patients who has disease type 1(psoriasis) and cluster 2 contains only disease type 3 patients (lichen planus). Cluster 1 contains the patients from five out of six disease types as summarized in table 4.

Main Attributes of each Clusters

Our data set mainly consists two set of features, that are clinical features (visible sings of the skin) and histopathological features (Internal skin patterns). Through descriptive analysis we found important unique features across these 3 clusters. Summary of them is given below table 5 and table 6.

✓ ***Unique Clinical symptom patterns***

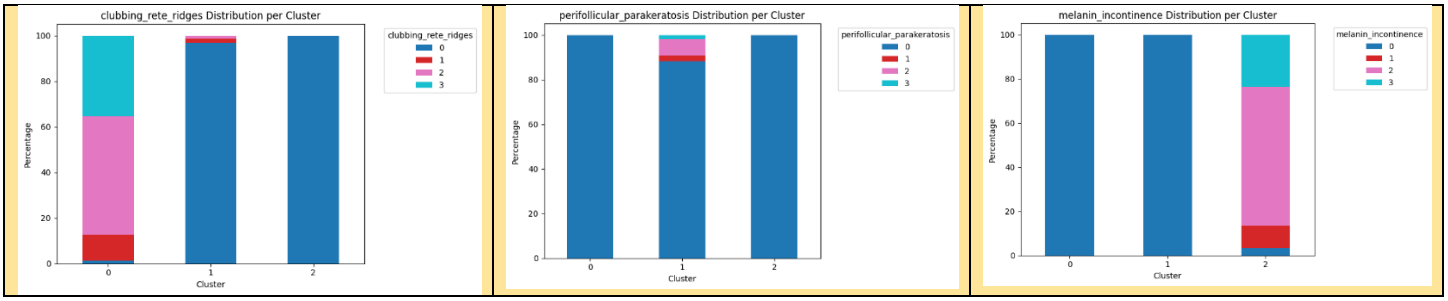
Table 5 : Unique Clinical features of each cluster

Cluster 0 – Psoriasis Cluster	Cluster 1- Mixed Cluster	Cluster 2 – Lichen Planus Cluster
• Knee and Elbow involvement • Scalp involvement	• Follicular papules	• Oral mucosal involvement • Polygonal papules
		

✓ ***Unique Histopathological Features***

Table 6: Unique Histopathological Features in each cluster

Cluster 0 – Psoriasis Cluster	Cluster 1- Mixed Cluster	Cluster 2 – Lichen Planus Cluster
• Clubbing of rete ridges • Thinning of suprapapillary epidermis • Spongiform pustule • Munro microabscess:	• Exocytosis • Perifollicular parakeratosis	• Melanin incontinence • Focal hypergranulosis • Vacuolization/damage of basal layer • Saw-tooth appearance of rete ridges • Band-like infiltrate



Main Results of the Advanced Analysis

As we explored in the descriptive analysis section, cluster 0 consists entirely of patients diagnosed with disease type 1 (psoriasis), while cluster 2 consists entirely of patients diagnosed with disease type 3 (lichen planus). This strong association suggests that simply identifying the cluster of a new observation is sufficient to accurately predict their disease type in these 2 clusters. However, cluster 1 presents a more complex scenario. It consists of patients from 5 out of 6 disease types, which are disease type1(psoriasis), type2(seboreic dermatitis), type4(pityriasis rosea), type5(chronic dermatitis) and type6(pityriasis rubra pilaris). As shown in Table 4, disease types 1 and 6 exhibit a clear class imbalance within this cluster. Therefore, when building machine learning models to predict disease type within Cluster 1, special care must be taken to address both the relatively small sample size (n = 154) and the class imbalance, to ensure robust and unbiased predictions.

In the descriptive analysis phase, K-means clustering was applied using FAMD scores to identify distinct groups within the data. Observations belonging to Cluster 1 were then extracted from the training set to simplify the machine learning model development process. Prior to model building, several key preprocessing steps were carried out. Ordinal variables were appropriately encoded, and One-Hot Encoding was applied to the binary attribute. To address class imbalance in the training data, the SMOTE oversampling technique was employed. The following machine learning models were then selected for application to the processed dataset.

1. **<u>Ridge Classifier and Lasso Model for Classification</u>**

- The ridge classifier is designed for multi-class classification tasks. It offers a distinct for classifying data points.
- The data set is scaled to apply ridge classifier.
- The implementation is done using sklearn.linear_model library
- The Lasso model, which stands for Least Absolute Shrinkage and Selection Operator, is a linear model that performs both classification and feature selection by applying L1 regularization. In our analysis, Lasso penalizes the absolute values of the model coefficients, effectively shrinking less important feature weights to zero.
- Implemented using LogisticRegression with penalty='l1' in scikit-learn
- Both Models has given similar results in training and test set after applying hyperparameter tuning by GridSearchCV.

	class	Precision	Recall	F1-score
Training set	2	0.94	0.96	0.95
	4	0.98	0.94	0.96
	5	0.99	0.99	0.99
	6	0.98	0.99	0.99
Accuracy on Training set				**0.976**
Testing set	2	0.9	0.99	0.95
	4	0.99	0.88	0.93
	5	0.99	0.99	0.99
	6	0.99	0.99	0.99
Accuracy on Testing set				**0.9667**

Table 7: Results of Ridge and Lasso classification models

2. **Naïve Bayes**
   - Since our data set has most of categorical distributed attributes, we applied Categorical Naïve Bayes classifier.
   - The data are sclaled by using MinMax scaler and applied CategoricalNB() from sklearn.naive_bayes

	class	Precision	Recall	F1-score
Training set	2	0.85	0.90	0.88
	4	0.90	0.85	0.87
	5	0.99	0.99	0.99
	6	0.98	0.99	0.99
Accuracy on Training set				**0.94**
Testing set	2	0.75	0.67	0.71
	4	0.88	0.88	0.88
	5	0.82	0.90	0.86
	6	0.99	0.99	0.99
Accuracy on Testing set				**0.83**

Table 8: Results of Naive Bayes

- The model performs well on the training set with high accuracy (0.94), but there's a noticeable drop in test accuracy (to 0.83), suggesting moderate overfitting.

3. **Decision Tree Classifier**

- Default parameters were used because they yielded the best accuracy and did not lead to overfitting.

	class	Precision	Recall	F1-score
Training set	2	0.99	0.99	0.99
	4	0.99	0.99	0.99
	5	0.99	0.99	0.99
	6	0.99	0.99	0.99
Accuracy on Training set				**0.99**
Testing set	2	0.99	1.00	0.95
	4	1.00	0.90	0.94
	5	1.00	1.00	1.00
	6	1.00	1.00	1.00
Accuracy on Testing set				**0.98**

Table 9 : Results of Decision Tree Classifier

4. **Random Forest Classifier**

- GridSearchCV was done because its default parameters lead to overfitting.
  Best Parameters: {'max_depth': 3, 'max_features': 'sqrt', 'min_samples_split': 10, 'n_estimators': 50}

	class	Precision	Recall	F1-score
Training set	2	0.90	1.00	0.95
	4	1.00	0.88	0.93
	5	1.00	1.00	1.00
	6	1.00	1.00	1.00
Accuracy on Training set				**0.97**
Testing set	2	0.90	0.98	0.94
	4	0.97	0.85	0.91
	5	1.00	1.00	1.00
	6	1.00	1.00	1.00
Accuracy on Testing set				**0.95**

Table 10: Results of Random Forest Classifier

5. **SVM**

- GridSearchCV was done to select best parameters.
- Best Hyperparameters: {'C': 10, 'gamma': 'scale', 'kernel': 'linear'}

	class	Precision	Recall	F1-score
Training set	2	1.00	1.00	1.00
	4	1.00	1.00	1.00
	5	1.00	1.00	1.00
	6	1.00	1.00	1.00
Accuracy on Training set				**1.00**
Testing set	2	0.90	1.00	0.95
	4	1.00	0.88	0.93
	5	1.00	1.00	1.00
	6	1.00	1.00	1.00
Accuracy on Testing set				**0.97**

Table 11: Results of SVM model

Best model:

Decision Tree (Accuracy: 98%, not overfitting)

Applied in-built feature selection method for decision tree and the model was refit with the 10 most important features. However, the accuracy remained the same.

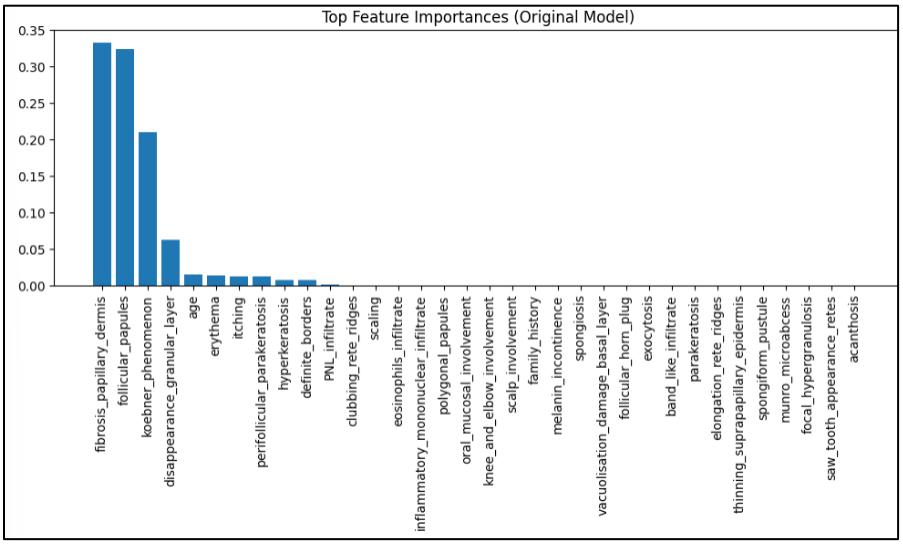
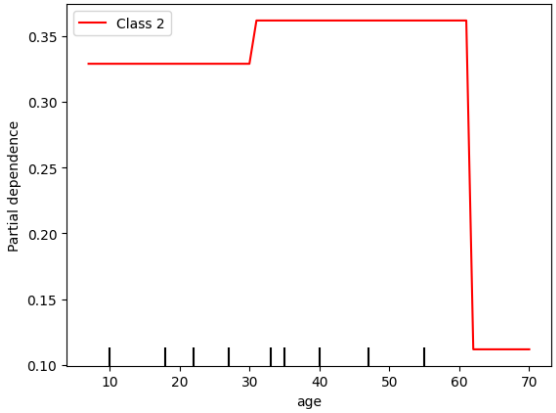


Figure 7: Top Feature Importances of Best model

Partial dependency plots of decision tree for age variable show some important patterns in disease class 2 and class 4 as below.

The model predicts a higher probability for Class 2 in the age range of ~30 to 60, with the highest response between those ages. After age 60, the partial dependence drops sharply.

The Partial Dependence Plot shows that for Class 4, the model's prediction stays constant across most age values, but jumps sharply after age 60, suggesting older age significantly increases the predicted probability for Class 4



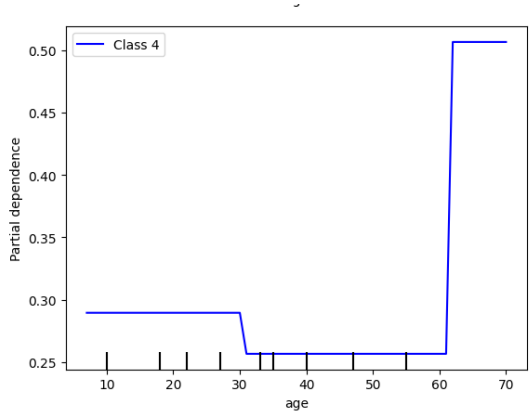


Figure 8: PDP plot of best model for Age variable

Issued we encounter and solutions

Cluster 1 contained 4 outliers detected via Isolation Forest, so we had to consider it during model building. Due to class imbalance, we applied SMOTE for oversampling and used tree-based models (Decision Tree, Random Forest) with class_weight='balanced'. Given the small sample size and high dimensionality, we prioritized simple models: regularized regression (Lasso/Ridge), SVM (with RBF kernel), and tree-based ensembles. Dimensionality reduction (FAMD) was explored but discarded as it explained only <30% variance with first 5 components. In our training dataset, Cluster 1 contained only 2 observations for disease class 1, which was insufficient for effective model training. To ensure consistency, we also removed the 4 observations of disease class 1 from Cluster 1 in the test set.

Discussion and Conclusion

This study utilized a comprehensive combination of descriptive, clustering, and machine learning methodology for the description of erythemato-squamous diseases (ESDs) based on clinical and histopathological features. The descriptive and cluster-based analyses revealed distinctive patterns among the six disease classes, in particular illustrating strong separability for lichen planus (Class 3) and psoriasis (Class 1), which were both separated into individual clusters. However, Cluster 1 had five disease types with mixed combination and was therefore the hardest and most clinically significant subset to model. We had performed focused preprocessing in the form of categorical feature encoding, applying SMOTE to handle class imbalance, and keeping Cluster 1 aside for advanced modeling. We experimented with various machine learning models—Naive Bayes, Ridge, Lasso, Random Forest, Support Vector Machine (SVM), and Decision Tree classifiers—to verify their predictive performance on this subset.

The highest performance was obtained using the Decision Tree classifier with 99% accuracy from the test set with a low amount of overfitting. Its ability to be interpreted and ability to model nonlinear relationships made it very effective in this multi-class classification setting. Additionally, constructing in the features with the top 10 variables did not decrease the performance, demonstrating that the model is stable and very efficient.

In conclusion, this research demonstrated that interpretable machine learning models, here decision trees, can effectively aid differential diagnosis of ESDs. The findings are informative about disease patterns and predictors and pave the way for more accurate and data-driven diagnostic tools in dermatology.

Appendix and References

Code Link: https://github.com/thamodya2001/Group-4-Data-Analysis-Project-2-

References:

1. https://scikit-learn.org/stable/modules/naive_bayes.html#categorical-naive-bayes
2. https://arxiv.org/pdf/1501.00607
3. https://physionyx.com/articles/mastering-machine-learning-limited-data